

Eighteen commits, three workstreams

What landed on `main` from `ce8ec31` through `c68d7e3`

FROM	<code>ce8ec312054ee238182f8c1776b5d7ad36f8b26b</code>
TO	<code>c68d7e33ca6e0cf274469279d52b3a1960923e99</code>
WINDOW	2026-09-16 23:47 → 2026-09-18 14:36 (Asia/Jerusalem)
COMMITTS	18 (13 direct, 5 merges)
NET CHANGE	69 files · +14,148 / -233 lines

1 Scope and how to read this

This report covers **18 commits**: `ce8ec31` itself and everything reachable from `main` that was not already reachable from `ce8ec31` 's parent. Every commit is included, merges among them.

One thing to be careful about: graph order is not chronological order.

`ce8ec31` was authored on the feature branch `EA-343-Resources-reach-the-most-critical-incident-first` at 22:12 on 09-17. Six commits in this range carry *earlier* timestamps — they were authored on `EA-337-Fire-events` between 09-16 23:47 and 09-17 18:20, and only became reachable from `ce8ec31` when the branches were merged afterwards.

So "commits after `ce8ec31` " in the Git sense includes work that was written *before* it. The table in §3 is sorted by author date to make this legible. Nothing authored by `YonatanReich` post-dates `ce8ec31` in wall-clock time.

Reading the line counts

Merge commits report a *combined* diff against their first parent, so their totals restate work already counted in the individual commits they integrate. The five merges in this range account for most of the apparent volume. The honest figure for net new content on `main` is the **69 files / +14,148 / -233** on the cover, and even that is dominated by reference data rather than code — see §2.

2 At a glance

18 COMMITTS	3 CONTRIBUTORS	69 FILES TOUCHED	14,148 LINES ADDED	3 MIGRATIONS
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Contributors

AUTHOR	COMMITTS	PRIMARY FOCUS IN THIS WINDOW
NaomiMeisler	9	Resource allocation — frontend presentation, station popups, route rendering, scheduler wiring
YonatanReich	6	Fire districts layer, searchable towns, repository hygiene, Alembic merge revision
idocohen11	3	Air pollution official classification, DB-backed localities, towns-layer integration

Where the lines went

AREA	FILES	LINES ADDED	CHARACTER OF THE CHANGE
<code>data/reference/</code>	19	+8,669	CBS population tables, authority contacts, police-station mapping, GeoJSON outlines
<code>frontend/</code>	10	+1,424	Three new map layers, town search, event modal rework
<code>scripts/</code>	3	+1,235	Reference-data build pipeline (towns, localities, municipal GeoJSON)
root & config	13	+971	ML artifacts, <code>.gitignore</code> , <code>.env.example</code> , README

AREA	FILES	LINES ADDED	CHARACTER OF THE CHANGE
tests/	10	+792	Classification, verification, migration and scheduler coverage
analyzers/	9	+715	Air pollution qualification, verification, trend inference
api/	3	+306	Two town endpoints, event payload extensions
database/	5	+524	towns table + repository, two merge revisions
coordinator/	2	+36	Projection plumbing for new event fields

Reference data is 61% of the added lines. The actual code change is roughly 5,000 lines.

3 Commit timeline

Sorted by author date. **MERGE** marks an integration commit whose stats restate work counted elsewhere.

COMMIT	DATE & TIME	AUTHOR	SUBJECT	FILES	+/-
4f15922	09-16 23:47	YonatanReich	Fire districts layer map	11	+282
62ab5be	09-17 11:41	idocohen11	EA-371: official air pollution classification and verification	18	+1239 -149
ce35f8c	09-17 12:28	idocohen11	Add DB-backed locality layer for air pollution	18	+941 -67
0437848	09-17 17:46	YonatanReich	Working searchable towns	36	+10967 -12
14c39b7	09-17 18:02	YonatanReich	Untrack GIS formats and scratch logs nothing reads	16	+11 -853
472e630	09-17 18:02	YonatanReich	Merge origin/main into EA-337-Fire-events MERGE	105	+17315 -16
332b8f9	09-17 18:11	YonatanReich	Add Alembic merge revision, deterministic locality ids	3	+37 -2
2730aa1	09-17 18:20	Yonatan Reich	Merge PR #31 — EA-337-Fire-events MERGE	72	+18654 -92
ce8ec31	09-17 22:12	NaomiMeisler	Present resource allocation in frontend ← start of range	9	+540 -3
7aad18e	09-17 22:31	NaomiMeisler	Add popup when pressing on a station	4	+240 -13
d96f3cc	09-17 22:36	NaomiMeisler	Add details about allocated stations	1	+16 -2
ca41682	09-17 22:47	NaomiMeisler	Improve routes representation	1	+78 -46
1c7ecaa	09-17 23:06	NaomiMeisler	Update routing presentation	2	+10 -3
38b9eac	09-18 13:01	idocohen11	Integrate Air Pollution with shared towns layer MERGE	88	+14302 -591
e99e6fe	09-18 13:54	NaomiMeisler	Remove resource allocation demo from frontend	3	+35 -62
29ca2d4	09-18 13:58	NaomiMeisler	Merge main into EA-343 MERGE	64	+13194 -229
2dda8d9	09-18 14:34	NaomiMeisler	Add resource allocation to scheduler flow	3	+160
c68d7e3	09-18 14:36	NaomiMeisler	Merge PR #32 — EA-343 MERGE	26	+5514 -359

4 What changed, by workstream

The 18 commits resolve into three feature workstreams and one housekeeping thread. They were developed in parallel on separate branches and converged through PR #31 and PR #32.

4.1 Fire districts and fire events FIRE

Fire districts layer map 4f15922

YonatanReich · 2026-09-16 23:47 · 11 files, +282

Adds the seven districts (מחוזות) of Israel's National Fire and Rescue Authority as a toggleable map layer, answering the operational question *which authority owns the ground this event landed on*.

Data. The source is `data/reference/Stations_border.shp`, a 925 KB shapefile carrying 34 polygons at *area* granularity. It is dissolved by district into `fire_districts.geojson`, collapsing the internal area boundaries so only the seven district outlines remain; the constituent area names survive as an `areas` property on each feature.

Rendering. `FireDistrictsLayer.tsx` (241 lines) bundles that GeoJSON at build time rather than fetching it, on the reasoning that district boundaries change about once a decade and a map redraw should never block on a network round trip for static reference geometry. Hovering a district raises its tint to solid and lights its border, so an operator can isolate one of seven overlapping fills without clicking. `LayersControl.tsx` gains the toggle; `Dashboard.tsx` mounts the layer.

4.2 Air pollution: official classification and verification AIR

EA-371: official air pollution classification and verification 62ab5be

idocohen11 · 2026-09-17 11:41 · 18 files, +1,239 / -149

Moves air-pollution event publication from ad-hoc rules embedded in the API onto an explicit, tested classification boundary. This is the largest structural change of the three workstreams.

Four new analyzer modules:

- `official_classification.py` — types the Ministry of Environmental Protection's pollutant sub-index bands. Deliberately conservative: values that are `None`, boolean, non-numeric or non-finite fall through to an unclassified result rather than being coerced, so a sensor glitch cannot manufacture a band.
- `event_qualification.py` — the Path A / Path B qualification rule, lifted out of the API and given an Air-Pollution-specific home. Its docstring is explicit about what it refuses to do: it does not classify p95, health severity, or response priority.
- `additional_verification.py` — a second pass over already-qualified significant events, correlating them against recent FIRMS fire activity. It ships a hard-coded `LIMITATION` string stating that fire evidence indicates *possible source correlation only* — not causation, not an emission source, not a plume, not exposure.
- `incident_handler.py` — the dispatcher-facing entry point for the pollution hazard.

Consequence for the API. `api/events.py` shrinks by 149 lines as the publication logic moves behind that boundary. `shared/events.py` and `frontend/src/types/events.ts` gain matching fields so the verification verdict and its limitation reach the UI, where `EventCard.tsx` and `EventModal.tsx` render it.

Tests. 199 new lines across `test_official_classification.py` and `test_additional_verification.py`, plus 87 lines added to the incident-dispatcher suite.

DB-backed locality layer for air pollution ce35f8c

idocohen11 · 2026-09-17 12:28 · 18 files, +941 / -67

Replaces file-based locality lookup with a database table, so spatial enrichment of a pollution event resolves affected settlements through Postgres rather than by re-reading GeoJSON.

- Migration `0016_localities.py` plus `repositories/localities.py` (132 lines).
- `detectors/air_pollution/spatial_enrichment.py` substantially reworked (+188/-67) to query the new table; `spatial_schemas.py` gains the corresponding typed shapes.
- `scripts/load_localities.py` (200 lines) ingests the reference data.
- Test coverage: migration, repository, loader, and 90 new lines in the spatial-correlation suite.

This migration was later superseded — see §5 on how `localities` and `towns` converged.

4.3 Searchable towns FIRE

Working searchable towns 0437848

YonatanReich · 2026-09-17 17:46 · 36 files, +10,967 / -12

The single largest commit in the range, though ~9,000 of those lines are reference data. It gives the operator a search box that finds any Israeli settlement by name and frames it on the map with its outline and details.

Reference data ingested. `authorities.json` (5,591 lines — local authority contacts), `population-and-households-by-locality.csv` (1,223 lines, CBS), `settlementsbypolicestations.csv` (1,325 lines — the settlement → responsible police station mapping), plus `towns.json`, `muni_il.geojson` and `localities_osm.geojson` as single-line compacted GeoJSON.

Build pipeline. Three scripts — `build_towns.py` (596 lines), `build_localities.py` (376) and `build_muni_geojson.py` (263) — turn the raw Ministry of Interior shapefile and CBS tables into the shipped artifacts.

Storage. Migration `0014_towns.py` creates the `towns` table with four indexes tuned to the access pattern: a GIST index on the outline geometry, two `text_pattern_ops` prefix indexes (Hebrew and lowercased English names) and one on `fire_district`. `repositories/towns.py` (244 lines) reads it.

Ranked, not filtered. The repository docstring is worth quoting in substance: an operator typing `בית` means one of roughly forty places and wants the obvious one first. So a prefix match outranks a mid-string match, and larger population outranks smaller. Returning forty rows in arbitrary order would be "technically correct and useless."

API. Two endpoints in `api/main.py`: `GET /api/towns/search` and `GET /api/towns/{town_id}`.

UI. `TownSearch.tsx` (306 lines) deliberately issues *two* requests rather than one. Search returns no geometry — an operator may type six characters before choosing, and shipping polygons on every keystroke to populate a dropdown would be wasteful. The outline is fetched only once a town is actually selected; the bounding box rides along with the search result so the camera can begin moving before the second request lands. The details card is hover-to-preview, click-to-pin.

`analyzers/emergency/fire/exposure.py` is reworked (+79) to consume the new table.

4.4 Air pollution meets the shared towns layer AIR

Integrate Air Pollution with shared towns layer 38b9eac MERGE

idocohen11 · 2026-09-18 13:01 · 88 files, +14,302 / -591

A merge of the fire-events line (2730aa1) with the air-pollution locality line (ce35f8c), resolving the two independent settlement models onto one shared `towns` layer. Beyond the merge itself it carries substantial new material:

- **Trend inference.** `trend_inference_service.py` (354 lines) performs live inference from approved SGD model artifacts, with four trained models shipped under `data/generated/ml/air_pollution_trend/final/` — `no2`, `o3`, `pm2_5` and `so2` `.joblib` files behind a 49 KB `manifest.json`. The manifest pins artifact version, feature-schema SHA-256, preprocessing version, embargo window and minimum history coverage, so a model cannot be served against features it was not trained on.
- **Generic dispatch.** `coordinator/dispatcher.py` (228 lines) introduces hazard-agnostic post-Coordinator analysis dispatch — a handler registry keyed by (hazard, route), with a typed `IncidentProcessingResult` and explicit success/partial/failed/skipped status.
- **Durable projections.** `coordinator/event_projection.py` grows to 451 lines, mapping processing results to persistent frontend-compatible `SharedEvent` records in the `event_projections` table.
- **Frontend.** `AirPollutionCorridorLayer.tsx` (78 lines) draws the predicted transport corridor and its centreline. Every emitted feature carries a `semantics` property reading `possible_transport_screening_not_confirmed_plume_or_exposure` — the epistemic caveat travels with the geometry rather than living only in a caption. `EventModal.tsx` is reworked (+489) and `Dashboard.tsx` restructured (+251).

4.5 Resource allocation ALLOCATION

Nine commits by NaomiMeisler, taking resource allocation from a frontend demo to a scheduler-driven pipeline backed by real routing.

Present resource allocation in frontend ce8ec31

NaomiMeisler · 2026-09-17 22:12 · 9 files, +540 / -3 · *range start*

First visualisation of which stations are assigned to an incident. `ResourceAllocationLayer.tsx` (80 lines at this point) plots assigned stations by type — fire (תחנת כיבוי, red `F`), police (תחנת משטרה, blue `P`), MDA (תחנת א"מ, green `M`).

Crucially it ships `api/resource_allocation_demo.py` (196 lines) and a `?include_resource_allocation_demo=true` query parameter, with the Dashboard gating on `import.meta.env.DEV`. The stated reason in the code comment: loading the preview automatically "*avoids the paid fire analysis endpoint during UI work.*" This scaffold is removed 16 hours later in e99e6fe — see below, because it explains a lot.

Station popups, details, and route rendering 7aad18e · d96f3cc · ca41682 · 1c7ecaa

NaomiMeisler · 2026-09-17 22:31 → 23:06 · 344 lines net across 4 commits

Four rapid iterations on the same surface, 35 minutes end to end:

- 7aad18e — clicking a station opens a popup; 109 lines of supporting `dashboard.css`.
- d96f3cc — the popup gains allocation detail (which unit, what role).
- ca41682 — route rendering substantially rewritten (+78/-46): the largest single revision of the layer.
- 1c7ecaa — final presentational pass on routing in `EventModal.tsx`.

Remove resource allocation demo from frontend e99e6fe

NaomiMeisler · 2026-09-18 13:54 · 3 files, +35 / -62

The most consequential small commit in the range, and the one most likely to confuse anyone pulling it. It deletes:

- the `RESOURCE_ALLOCATION_DEMO = import.meta.env.DEV` constant,
- the conditional that skipped `/api/detected-events` in development,
- the `?include_resource_allocation_demo=true` branch, and its 18 lines of server-side support in `api/events.py`.

Both fetches become unconditional: `fetch('/api/detected-events')` and `fetch('/api/events')`.

Two effects on a freshly-pulled checkout. First, the allocation layer now renders real data — and renders nothing at all if `event_projections` and `resource_allocations` are empty, which they are until the scheduler has run. Second, every Dashboard mount now calls the full detection pipeline, including two LLM reasoning agents. The guard that used to prevent that during UI work is gone.

Add resource allocation to scheduler flow 2dda8d9

NaomiMeisler · 2026-09-18 14:34 · 3 files, +160

Wires allocation into the recurring pipeline — the replacement for the demo feed. `scheduler.py` gains `allocate_resources(processing_results)`, which filters to fire-hazard results on the emergency route holding a response plan, then issues **one batched call** to `resource_allocator.allocate_batch()`.

The batching is not an optimisation, and the code says so: the allocator must compare incidents that compete for the same stations *before* making any assignment. Allocating incident-by-incident would let an early low-priority event claim a station a later critical one needs — which is exactly the ticket title, "*resources reach the most critical incident first.*" Results are mapped back onto each `IncidentProcessingResult` by incident id. 101 lines of scheduler tests accompany it.

Merge PR #32 — EA-343 c68d7e3 MERGE

NaomiMeisler · 2026-09-18 14:36 · 26 files, +5,514 / -359 · current HEAD

Brings the allocator engine itself onto `main`:

- `resource_allocator/allocation_agent.py` — heavily reworked (866 lines changed). Station requirements scale by risk band (low → 1/1/1 through critical → 4/2/2 for fire/police/medical). Candidate stations are first ranked by haversine distance, then re-ranked by *road* distance, then claimed transactionally.
- `resource_allocator/mapbox_client.py` (342 lines) — a synchronous client for the Mapbox Matrix and Directions APIs, with `RoutingError` and `RouteNotFoundError` distinguishing "routing is down" from "no road route exists", so the UI can say which.
- `repositories/resource_allocations.py` (231 lines) — transactional claim and release, with a release history rather than destructive updates.
- Migration `0016_resource_allocations.py` (83 lines).
- Tests: `test_batch_allocation.py` (676 lines) and `test_mapbox_client.py` (167) replace the older 199-line agent suite.

4.6 Repository housekeeping UPKEEP

Untrack GIS formats and scratch logs nothing reads 14c39b7

YonatanReich · 2026-09-17 18:02 · 16 files, +11 / -853

Removes **32 MB** that every clone would otherwise carry permanently: `muni_il.dwg` (16.9 MB, an AutoCAD copy) and `muni_il.zip` (15.1 MB, a zipped duplicate) — both redundant renderings of the same Ministry of Interior layer, neither referenced by any code. Also eight leftover ArcGIS `.sr.lock` session files and five `fire_weather_*.txt` replay scratch outputs (853 lines).

The shapefile itself stays: `scripts/build_muni_geojson.py` reads it to produce `muni_il.geojson`, and the commit message records that regeneration is byte-identical without the removed files. `.gitignore` gains 11 lines to keep them out.

Alembic merge revision and deterministic locality ids 332b8f9

YonatanReich · 2026-09-17 18:11 · 3 files, +37 / -2

Two unrelated fixes sharing a commit.

Alembic. Merging main left two heads — `air_pollution_weather_merge` and `towns`, both descended from `weather_baselines` — and `alembic upgrade head` refuses to choose. Revision 0015 joins them, mirroring what 0012 did for the previous pair.

Determinism. `build_localities.py` was using `id(tags)` as a fallback locality id. That is a Python memory address: 127 ids changed on every run and the output file never reproduced itself. Hashing the Hebrew name instead makes rebuilds byte-identical. `towns.json` was unaffected — its ids come from CBS codes.

Integration merges 472e630 · 2730aa1 · 29ca2d4

YonatanReich, Yonatan Reich, NaomiMeisler · 09-17 18:02 → 09-18 13:58

Three integration points carrying no original content. `472e630` pulls main into `EA-337-Fire-events`; `2730aa1` merges that branch via PR #31; `29ca2d4` pulls the result into `EA-343` ahead of PR #32. Their large line counts restate work already attributed above.

5 Database changes

Three migrations reached `main` in this range, two of them merge revisions that create no schema:

REVISION	FILE	EFFECT
towns	0014_towns.py	Creates the <code>towns</code> table (206 lines) — GIST index on outline geometry, Hebrew and English prefix indexes, <code>fire_district</code> index
towns_air_pollution_merge	0015_merge_towns_air_pollution_heads.py	Joins <code>air_pollution_weather_merge</code> + <code>towns</code> . No schema operations
event_projections_towns_merge	0016_merge_event_projections_towns_heads.py	Joins <code>event_projections</code> + <code>towns_air_pollution_merge</code> . No schema operations

A dedicated test, `test_event_projections_towns_migration_merge.py`, statically parses the 0016 revision's AST to assert it contains no `Call` nodes in either `upgrade()` or `downgrade()` — i.e. that the merge revision really is inert and did not silently acquire a `CREATE TABLE localities`. The test never connects to a database.

Migration graph state

`main` currently has **two heads**, both descending from `event_projections`:



They need an `alembic merge` to return to a single head. See §7 for the related operational defect.

6 API and frontend surface

New endpoints

ENDPOINT	ADDED IN	PURPOSE
GET /api/towns/search	0437848	Ranked name search; returns metadata and bounding box, no geometry
GET /api/towns/{town_id}	0437848	Full record including outline polygon, population, fire district, police station, authority contact

Removed surface

?include_resource_allocation_demo=true on /api/events , removed in e99e6fe together with api/resource_allocation_demo.py 's wiring. The only remaining reference anywhere in the tree is a commented-out line at tests/api/test_events.py:179 .

New frontend components

COMPONENT	LINES	ROLE
TownSearch.tsx	306	Settlement search, camera framing, outline and details card
layers/FireDistrictsLayer.tsx	241	Seven fire & rescue districts, hover-to-isolate
layers/ResourceAllocationLayer.tsx	189	Allocated stations, routes, click-through popups
layers/AirPollutionCorridorLayer.tsx	78	Predicted transport corridor and centreline

Substantially reworked: EventModal.tsx (+489 across the range), Dashboard.tsx (+251), dashboard.css (+220 over four commits), types/events.ts (+340), EventCard.tsx , LayersControl.tsx , hazards.ts , MapView.tsx .

Test coverage added

792 lines across 10 files — the largest being test_batch_allocation.py (676, arriving via PR #32), test_additional_verification.py (129), test_spatial_correlation.py (+110), test_events.py (+108), test_towns_nearby_repository.py (105) and test_scheduler_detection.py (101).

7 Operational notes for anyone pulling this range

Three conditions were observed on a checkout at `c68d7e3` against the shared Neon database. None is caused by the commits themselves, but all three shape what you see after pulling.

1. The feature tables are empty, so the new layers render nothing.

`event_projections` : 0 rows. `resource_allocations` : 0 rows. (`towns` is populated — 1,168 rows.) Because `e99e6fe` removed the demo feed, `ResourceAllocationLayer` and `AirPollutionCorridorLayer` now mount against real data and draw nothing until the coordinator has run. The code is present and correct; the data is absent. Populate via `detect_and_coordinate()` in `scheduler.py`.

2. `alembic_version` points at a revision that does not exist.

The shared database's `alembic_version` table holds `flood_incident_lifecycle`. A pickaxe search across every commit on every branch finds no such revision — it was stamped from a migration file that was never committed, then lost with its branch. Consequences: `alembic current` errors, and `alembic upgrade` cannot run.

The schema itself is *correct* — every table and column the `main` DAG defines was verified present, including the `ALTER`-style additions on `observations`. Only the marker is wrong. The repair is `alembic stamp --purge heads` (`--purge` avoids resolving the unlocatable current id; `heads` writes both). Because the database is shared, this should be coordinated with the team rather than applied unilaterally. Nothing at runtime reads `alembic_version`, so deferring is safe — it bites only the next person to add a migration.

3. The Dashboard now bills on every page load.

`/api/detected-events` runs the full pipeline — `FireDetectionAgent`, then `RiskAnalysisAgent` and `ResponsePlanningAgent`, both LLM-backed. The `import.meta.env.DEV` guard that suppressed this during UI work was removed in `e99e6fe` and not replaced. The endpoint accepts `include_analysis=false` to skip both model calls; the Dashboard does not currently pass it.

Suggested follow-ups

1. Agree a policy that feature-branch migrations are not applied directly to the shared Neon database — that practice is the root cause of note 2, and the stamp only clears the symptom.
2. Add an `alembic merge` revision to collapse the two current heads.
3. Decide whether the Dashboard should pass `include_analysis=false` by default in development, restoring the cost guard that `e99e6fe` removed without reintroducing the synthetic demo payload.
4. Seed `event_projections` so a fresh clone can exercise the allocation and corridor layers without a paid pipeline run.